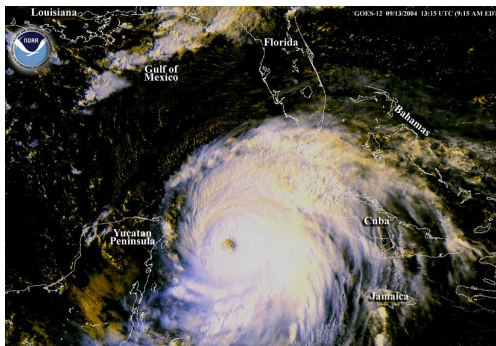


Detection of changes in variance of oceanographic time-series using changepoint analysis

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- Motivation and background to GOMOS dataset
- Changepoint methodology
- Simulation Study
- Application to GOMOS data
- Conclusion

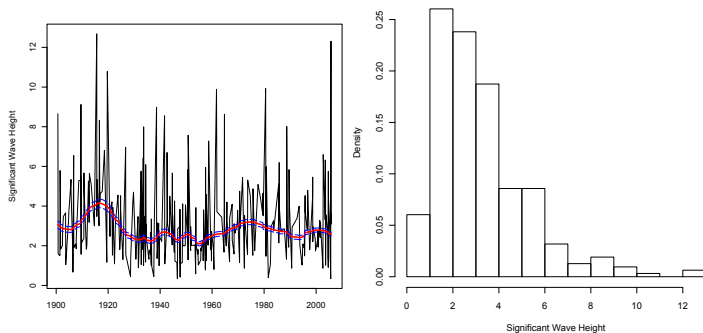
Motivation

- Hindcasts are used in offshore design
- Cooper & Stear (2006) suggest early 20th century storms are underestimated
- Partial reasoning: data collection methods have changed
 - Pre-1944: Coastal estimates
 - 1945-1960: Flying into storms
 - 1960-1990: Improved equipment on the airplanes
 - Post-1990: More modernization of equipment
- Recent storms (Ivan, Katrina), may not be unusual
- Authors suggest a change in variability may have occurred

Consequently: some analysts ignore pre-1950 data

Gulf Of Mexico Oceanographic Survey

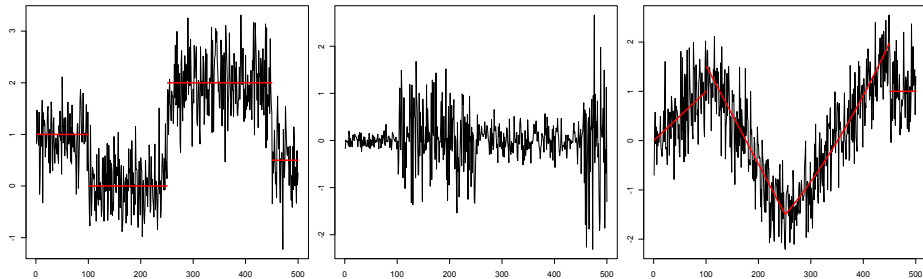
- 2658 locations in the Gulf of Mexico
- 315 hurricane storms over 1900–2005
- All locations have similar characteristics



Changepoint Methodology

What is a changepoint?

For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.



Changes at 100, 250, 450.

Change in Variance

Typical Problem Setup

For data $y_1, \dots, y_n$,

$$H_0 : \quad \sigma_1 = \dots = \sigma_n = \sigma$$

$$H_1 : \quad \sigma_1 = \dots = \sigma_\tau \neq \sigma_{\tau+1} = \dots = \sigma_n$$

The likelihood method test statistic is:

$$\Lambda(y_1, \dots, y_n) = -2 \max_{\tau} \log \left(\frac{L(\hat{\sigma}, \hat{\Theta} | y_{1:n}, H_0)}{L(\hat{\sigma}_1, \hat{\sigma}_n, \hat{\Theta}, \tau | y_{1:n}, H_1)} \right).$$

The statistic is compared to a confidence limit to decide if there is evidence for a changepoint or not.

Assumptions of Normal LR method

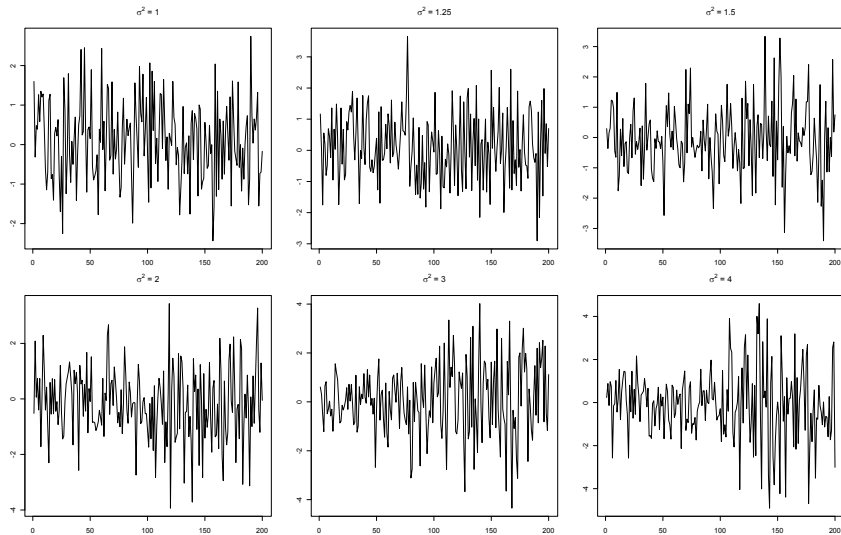
- Mean is constant
- Assumed distribution of data is correct (i.e. normal)
- Data are independent

Simulation Study

Simulation Study

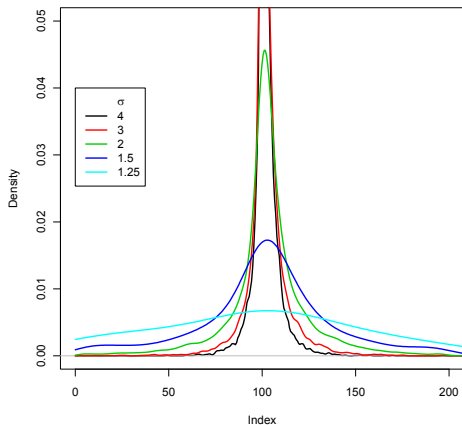
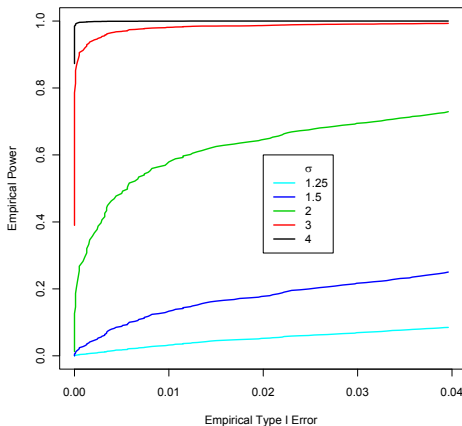
- First 100 data points from $N(0,1)$
- Next 100 data points from $N(0,\sigma^2)$
- 6 Scenarios, $\sigma^2=1, 1.25, 1.5, 2, 3, 4$.
- Simulate 10,000 replications

Simulation Example



Simulation Results

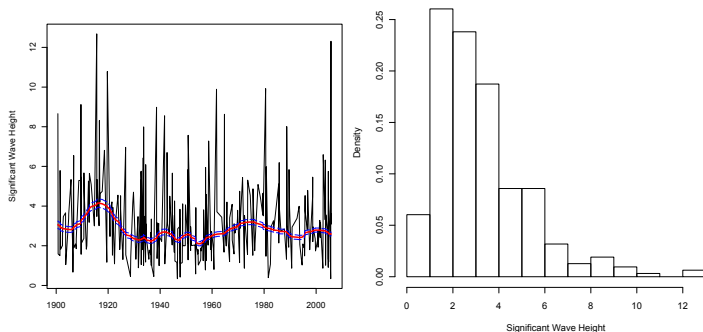
Cannot accurately detect a variance ratio less than 2.



Analysis of GOMOS data

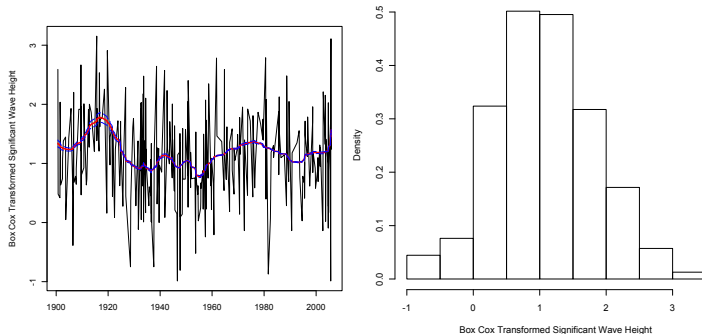
GOMOS Analysis

Take each location and analyse them separately.



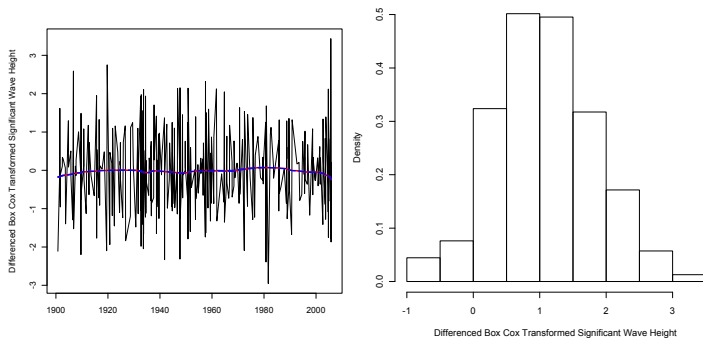
The histogram shows we do not have Normality ...

... so we perform a Box-Cox transformation.



But the mean is not constant, ...

... so we then take first differences.



Changepoint Analysis

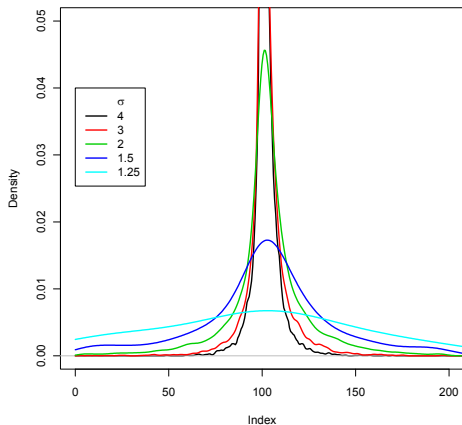
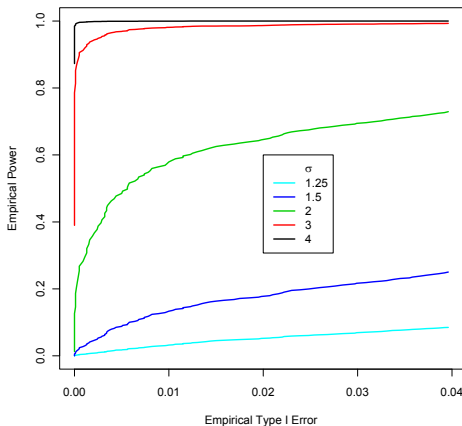
Now the assumptions are satisfied we run the changepoint algorithm on the transformed data.

First differenced Box-Cox Transformed Significant Wave Heights						
	No. of significant locations with storm at date (yyyy-mm)					
α	(1900-10)	(1905-10)	(1916-11)	(1925-09)	(1925-12)	(1933-09)
0.1	3	3	55	5	10	25
0.05	2	-	29	-	-	2
0.01	-	-	-	-	-	-

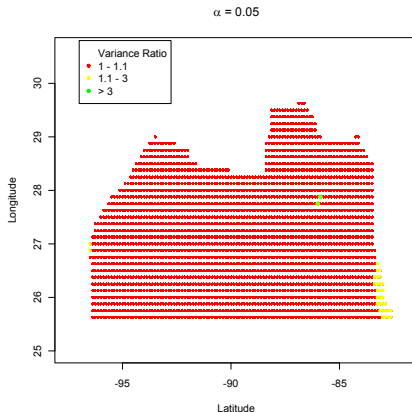
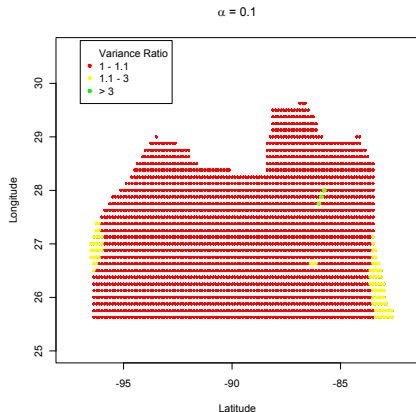
Number of locations (out of 2658) that found a change at date x .

Simulation Results

Cannot accurately detect a variance ratio less than 2.



Results



The two green points have variance ratios greater than 4000.

Conclusion

No evidence of changes in variance

Future work includes

- looking at the effect of transformations on changepoint algorithms
- incorporating spatial dependence
- monitoring changes in mean and variance simultaneously e.g. DLMs
- other distributional assumptions such as Weibull.

Any Questions?